清华大学能源与动力工程系 2020 春《燃烧学》期中考试
（A 卷）
Middle Term Examination of Combustion (A)

考试时间(Date)： 2020 年 4 月 21 日 班级(Class)：_____

学号(Student No.)：_____ 姓名(Name)：_____

提示 (Instructions):

1 请将你的学生证放在桌子的右上角（Please put your student ID on the desk）

2 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they cannot be lent to others. You should answer the questions independently and help is not allowed）

3 试卷共有**5**题，你要在**120**分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有**6**页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are **5** questions on this exam with **6** pages. You have up to **120** minutes to work on this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

4 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

5 强烈推荐你先读一下所有的问题，并对回答问题的时间进行安排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

6 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸首页上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages. Mark the exam type A or B）

7 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

一、简答题：（42 分）

1. 什么是分子扩散速度？针对二元扩散，尝试如何用浓度差来建立组分 A 的分子扩散速度 $V_{A,diff}$？(7 分)

What is the diffusive velocity of a molecular?. For binary diffusion, try to derive the diffusive velocity of molecular A $V_{A,diff}$. (7 points)

2．对于一个反应 $aA+bB+\ldots\ldots \Leftrightarrow eE+fF+\ldots\ldots$，其对应的吉布斯函数差 $\Delta G_T^0 = -R_u T \ln K_p$，有人认为是$\Delta G_T^0$是实际温度、混合物 1atm 下产物/反应物的吉布斯函数差，也有人认为是实际温度、每个组分都在 1atm 下产物/反应物的吉布斯函数差，你如何判断，并简要说说明。 (6 分)

For a chemical reaction $aA+bB+\ldots\ldots \Leftrightarrow eE+fF+\ldots\ldots$, the variation of Gibbs function $\Delta G_T^0 = -R_u T \ln K_p$, some people regard ΔG_T^0 as the variation of Gibbs function of the products/reactants under the actual temperature while the mixture is at 1 atm, some people regard ΔG_T^0 as the variation of Gibbs function of the products/reactants under the actual temperature while each single species is under 1atm. Please give your comments on this problem. (6 points)

3. 什么是爆炸极限？有人认为压力越高，氢气越容易发生爆炸，你认为对吗？并请简述原因。（8 分）

What is the definition of explosion limit? Some people think that the higher the pressure will lead to easier explosion of hydrogen, do you think it is right? Please briefly explain the reason. (8 points)

4. 将一滴墨水滴到水中，墨水会逐渐与水混合在一起。这种现象是什么？用什么定律描述？现在为了加快墨水和水的混合，以采用哪些方法？为什么？请举出 2 条措施(8 分)。

Drop a drop of ink into the water, and the ink will gradually mix with the water. What is this phenomenon? What law is used to describe this phenomenon? In order to speed up the mixing of ink and water, which methods can be taken? Why? Please give at least 2 methods (8 points).

5. 请问什么是守恒标量的定义？请列出至少三种守恒标量。（5 分）

What is the definition of conserved scalar? Please list at least three conserved scalars. (5 points)

6. 在燃烧过程中会生成许多高反应性自由基。在处理有自由基参与的反应时，我们常采用稳态近似和部分平衡两种方法。请问什么是稳态近似和部分平衡？并请具体各举出一个应用这两种处理方法的化学反应实际例子。(8 分)

During the combustion process, many highly reactive free radicals are generated. When dealing with reactions involving these free radicals, we often use steady-state approximation and partial equilibrium. What is the steady-state approximation and partial equilibrium? Please give suitable chemical reaction example that use these two treatment methods, respectively. (8 points)

(13 分)二、Zeldovich 机理 NO 的形成首先是氧分子离解成为氧原子：

$$\mathbf{O_2 + M \longleftrightarrow 2O + M} \quad (\mathbf{1})$$

然后是以下述反应形成 NO：

$$\mathbf{O + N_2 \longleftrightarrow NO + N} \quad (\mathbf{2})$$

$$\boldsymbol{k_{1f} = 1.82 \times 10^{14} \times \exp(-38370/T(\mathrm{K}))} \quad \mathbf{cm^3/(mol \cdot s)}$$

$$\mathbf{N + O_2 \longleftrightarrow NO + O} \quad (\mathbf{3})$$

$$\boldsymbol{k_{2f} = 6.4 \times 10^{9} \times \exp(-3150/T(\mathrm{K}))} \quad \mathbf{cm^3/(mol \cdot s)}$$

上述三个反应都是可逆的，并且 O 原子和 N 原子的浓度在燃烧过程中保持准稳态。

（1） 假设 NO 的浓度很低，反应（2）和（3）的逆反应可以忽略。试用稳态组分的浓度来表达出 NO 的生成速率。（6 分）

（2） 假设一开始空气的温度被加热到 3000K，试采用上式计算 NO 的初始生成速率。（5 分）

（3） 如果考虑上述反应的逆反应，（1）和（2）的结论有何变化？（2 分）

The first step of Zeldovich mechanism of NO formation is the decomposition of O_2

$$\mathrm{O_2 + M \longleftrightarrow 2O + M} \quad (1)$$

And then NO forms by the following reactions

$$\mathrm{O + N_2 \longleftrightarrow NO + N} \quad (2)$$

$$k_{1f} = 1.82 \times 10^{14} \times \exp(-38370/T(\mathrm{K})) \quad \mathrm{cm^3/(mol \cdot s)}$$

$$\mathrm{N + O_2 \longleftrightarrow NO + O} \quad (3)$$

$$k_{2f} = 6.4 \times 10^{9} \times \exp(-3150/T(\mathrm{K})) \quad \mathrm{cm^3/(mol \cdot s)}$$

These three reactions are reservable, and atom O and N are at quasi-stable state.

(1) Suppose that the concentration of NO is low, the reverse reaction of (2) and (3) can be neglected. Try to express the formation speed of NO with the concentration of stable species.

(2) Suppose that at first the air is heated up to 3000 K, try to compute the formation speed of NO with the former expression.

(3) If the reverse reactions are taken into consideration, how will the conclusion of (1) and (2) change?

(15 分)三、丙烷和空气在一个全混流反应器内反应，化学反应式可以写成：

$$\mathbf{C_3H_8 + 5(O_2 + 3.76N_2) \rightarrow 3CO_2 + 4H_2O + 18.8N_2}$$

在反应器入口通入当量比为 1，温度为 298 K 的预混气体。燃料的总包反应速率：

$$\mathbf{\frac{d[C_3H_8]}{dt} = -A\exp(-E_a/R_uT)[C_3H_8]^{0.1}[O_2]^{1.65}}$$

其中$E_a/R_u = 15098$ K，$A = 8.6 \times 10^{11}$ $(\mathrm{mol/cm^3})^{-0.75}/\mathrm{s}$，各组分均可视为理想气体。反应器容积$V = 100\ \mathrm{cm^3}$，内部压力为一个大气压，温度$T = 2200$ K，不考虑产物的分解。

（1） 为了使燃料转化率（反应消耗的燃料占总燃料比例）为 90%，通入气体总流量$\dot{m}$应该是多少？反应物的平均停留时间t_R是多少？（9 分）

（2） 如果将该反应器压力提高到 10 个大气压，其他条件不变。假设此时仍符合全混流反应器模型，采取哪种措施可以使燃料转化率依然是 90%？（6 分）

Propane reacts with air in a perfectly stirred reactor, the reaction can be expressed as

$$\mathrm{C_3H_8 + 5(O_2 + 3.76N_2) \rightarrow 3CO_2 + 4H_2O + 18.8N_2}$$

The inlet mixture is at an equivalence ratio of 1 and 298 K. The reaction rate is

$$\frac{\mathrm{d[C_3H_8]}}{\mathrm{d}t} = -A\exp(-E_a/R_uT)[\mathrm{C_3H_8}]^{0.1}[\mathrm{O_2}]^{1.65}$$

where $E_a/R_u = 15098$ K，$A = 8.6 \times 10^{11}$ $(\mathrm{mol/cm^3})^{-0.75}/\mathrm{s}$, the species can be considered as ideal gas. The volume of the reactor $V = 100\ \mathrm{cm^3}$, the pressure in the reactor is 1 atm, temperature $T = 2200$ K, decomposition of products can be neglected.

(1) In order to keep the fuel consuming ratio (the ratio of consumed fuel over total fuel) at 90%, what is the total flow rate $\dot{m}$? What is the residence time t_R?

(2) If the pressure is raised to 10 atm and other conditions remain constant, suppose that the reactor can also be considered as a perfectly stirred reactor, what operation should be taken to keep the fuel consuming ratio remaining 90%?

（14 分）四、正辛烷（C_8H_{18}，相对分子量 114.23）是工业用汽油成分之一，还可用作溶剂和有机合成原料。在 298 K 时，正辛烷的气化潜热为 41465.49 kJ/kmol，生成焓为 -208447 kJ/kmol。在实际中，为了正辛烷能够完全燃烧和捕集碳，采用过量的 30%O2-70%CO2 作为氧化剂，假定此时的当量比为 0.8。请问：

（1） 在此当量比下，燃烧能够达到的最高温度？（6 分）

（2） 如果将氧化剂替换为 50%-50%的 O2-CO2 混合物，并在当量比为 1 时燃烧。请问此时，燃烧能够达到的最高温度?比（1）中高还是低？为了调整到和（1）中相同，需要如何操作？请提出一种方法。（8 分）

注：最终计算结果保留两位小数；

物质	定压比热容（kJ/kmol）	标准生成焓（kJ/kmol）
O_2	**35.59**	**——**
CO_2	**56.21**	**-393546**
$H_2O(g)$	**43.87**	**-241847**

n-octane (C_8H_{18}, relative molecular weight 114.23) is one of the components of industrial gasoline, and can also be used as a solvent and organic synthesis raw materials. At 298 K, the latent heat of vaporization of n-octane is 41465.49 kJ / kmol, and the formation enthalpy is -208447 kJ / kmol. In practice, to completely burn n-octane and capture carbon, a mixture oxidant of 30% O_2-70% CO_2 is used. Assuming that the equivalent ratio at this time is 0.8. Please answer:

(1) Under this equivalence ratio, the highest temperature that combustion can reach? (6 points)

(2) If the oxidant is replaced with a 50% -50% O2-CO2 mixture，and the equivalence ratio is 1. At this condition, what is the highest temperature that combustion can be reached? Is it higher or lower than temperature in (1)? In order to adjust the temperature as the same as the temperature in (1), how to operate? Please suggest a method. (8 points)

Note: The final calculation result retains two decimal places;

	Specific pressure specific heat capacity（kJ/kmol）	Standard enthalpy of formation（kJ/kmol）
O_2	35.59	——
CO_2	56.21	-393546

（16 分）五、一个直径为 120 μm 的水滴，在温度 353K(~摄氏 70°C)空气环境中蒸发，请分别计算常压 1 atm 和高压 5 atm 条件下的水滴寿命。根据蒸气表，353K 对应的饱和蒸汽压为 47342 Pa。在 298 K、1atm 下，水蒸气在空气中的扩散系数为 2.6×10^{-5} m^2/s，水的密度为 1000 kg/m^3，相对分子量 18。空气的相对分子质量取 28.85。

A water droplet with a diameter of 120 μm evaporates in an air environment at a temperature of 353K (~ 70°C). Please calculate the evaporation time of the water droplet under the conditions of normal pressure 1 atm and high pressure 5 atm, respectively. According to the steam table, the saturated steam pressure corresponding to 353K is 47342 Pa. At 298 K and 1 atm, the diffusion coefficient of water vapor in air is 2.6×10^{-5} m^2 / s, the density of water is 1000 kg / m^3, and the relative molecular weight is 18. The relative molecular mass of air is 28.85 kg/kmol.